

PROGRAM:

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#include <stdio.h>

// Function for linear search

int linearSearch(int arr[], int n, int key) {

    for (int i = 0; i < n; ++i) {

        if (arr[i] == key) {

            return i;

        }

    }

    // Return -1 if element is not found

    return -1;

}

int main() {

    int arr[] = {12, 45, 67, 23, 56, 89, 9, 43};

    int n = sizeof(arr) / sizeof(arr[0]);

    int key, index;

    printf("\n—Linear Search--\n");

    printf("Enter the element to search: ");

    scanf("%d", &key);

    index = linearSearch(arr, n, key);

    if (index != -1) {

        printf("Element %d found at position %d.\n", key, index + 1);

    }

    else {

        printf("Element %d not found in the array.\n", key);

    }

}
```

```
}  
    return 0;  
}
```

OUTPUT:

—Linear Search--

Enter the element to search: 45

Element 45 found at position 2.

—Linear Search--

Enter the element to search: 11

Element 11 not found in the array.